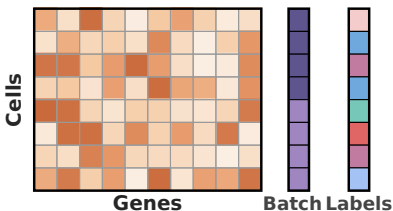
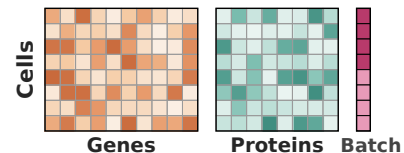


A Inputs

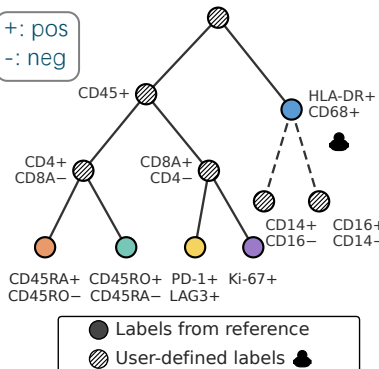
Reference (labeled)



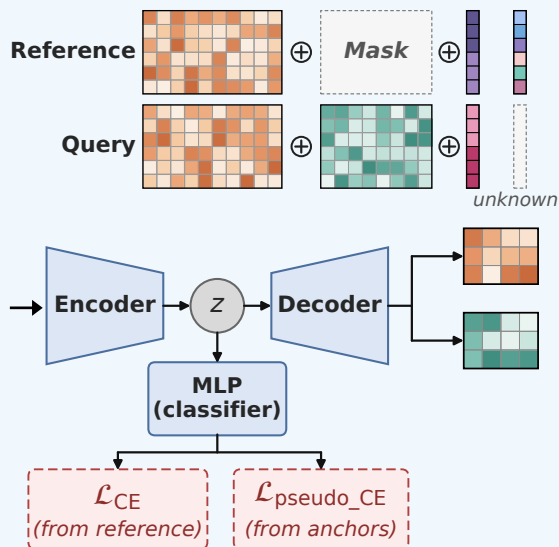
Query (RNA + Protein, unlabeled)



Hierarchical marker tree

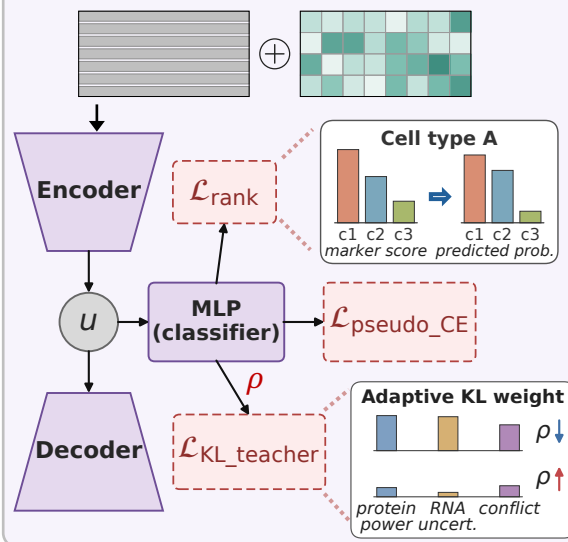


B Teacher model



D Student model

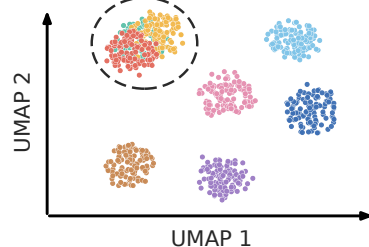
Input (Query only): $z \oplus \text{protein}$



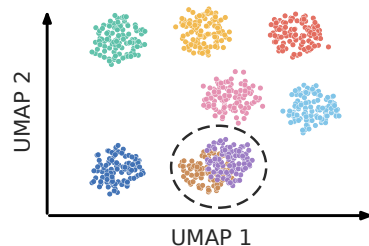
E Output:

Refined annotation

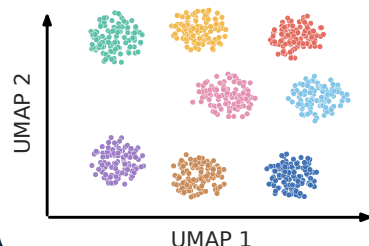
RNA only



Protein only

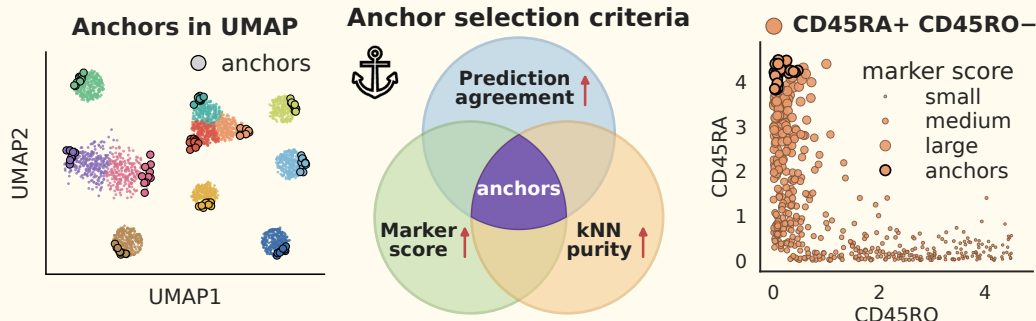


ANCHOR (RNA + Protein)



C Marker-guided anchor selection

Anchor selection criteria



$$\text{marker score} = \text{mean}(\text{pos}) - \text{mean}(\text{neg})$$